

JEE MAIN REVISION MASTER: CURRENT ELECTRICITY

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 66 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Ohm's Law, Drift Velocity & Resistance Scaling	54 PYQs	Conduction & Resistance
High	Kirchhoff's Laws, Bridges & Measuring Instruments	13 PYQs	Circuits & Instruments

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Ohm's Law, Drift Velocity & Resistance Scaling	$I = n e A v_d$, $R = \rho \frac{L}{A}$, $R \propto L^2$ $R \propto \frac{1}{r^4}$, $R' = n^2 R$
Kirchhoff's Laws, Bridges & Measuring Instruments	$\text{Meter Bridge: } R = S \frac{l}{100-l}$, $\text{Potentiometer: } \frac{E_1}{E_2} = \frac{l_1}{l_2}$, $r = R \left(\frac{l_1}{l_2} - 1 \right)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Ohm's Law, Drift Velocity & Resistance Scaling (Very High Yield — 54 PYQs)

Core Principle: Drift velocity $v_d = (e E \tau / m)$, current $I = n e A v_d$. Resistance $R = \rho (L / A) = \rho (L / \pi r^2)$. Temperature dependence $R_t = R_0(1 + \alpha \Delta T)$ ($\alpha > 0$ for metals, $\alpha < 0$ for semiconductors). When a wire of resistance R is stretched to n times its length at constant volume, its new resistance is $R' = n^2 R$.

■ **Caution / JEE Pitfall:** If length is stretched by $x\%$ ($x \leq 5\%$), resistance increases by $2x\%$! If radius is decreased by $x\%$, resistance increases by $4x\%$.

■ **Speed Trick:** Resistor color code mnemonic: 'BB ROY of Great Britain has a Very Good Wife' (Black 0, Brown 1, Red 2, Orange 3, Yellow 4, Green 5, Blue 6, Violet 7, Grey 8, White 9; Gold 5%, Silver 10%).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Given below are two statements: Statement-I: The resistivity of a conductor is independent of its temperature. Statement-II: The resistivity..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Which of the following circuits is correct for verification of Ohm's law?..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "In a conducting region, 10^{19} electrons and 10^{19} protons move to the left, while $10^{19} \alpha$ -particles move to the right per..." [Ans: B]

3.2 Kirchhoff's Laws, Bridges & Measuring Instruments (High Yield — 13 PYQs)

Core Principle: Kirchhoff's Current Law (junction rule: charge conservation) & Voltage Law (loop rule: energy conservation). Balanced Wheatstone bridge $(P / Q) = (R / S)$. Meter bridge: unknown resistance $R = S \left(\frac{l}{100-l} \right)$. Potentiometer draws no current at null point: $(E_1 / E_2) = (l_1 / l_2)$; internal resistance $r = R \left(\frac{l_1}{l_2} - 1 \right)$.

■ **Caution / JEE Pitfall:** In a balanced Wheatstone bridge, the galvanometer resistance has ZERO effect on the circuit and can be removed completely.

■ **Speed Trick:** Potentiometer is preferred over a voltmeter for measuring EMF because it measures open-circuit potential without drawing any current (infinite effective resistance).

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "In the figure, the values of currents I_1, I_2 and I_3 respectively are..." **[Ans: N/A]**
- **[2026 Shift] Q#2 (MCQ):** "The number of electrons moving per second through the filament of a lamp of 60 W operating at 120 V is nearly $(e = 1.6 \times 10^{-19})$..." **[Ans: C]**
- **[2024 Shift] Q#4 (MCQ):** "In the circuit shown, the end A is at potential V_0 and end B is grounded. The electric current I indicated in the circuit is..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In the figure, the values of currents I_1, I_2 and I_3 respectively are

- (A) 6 A, 1.5 A and 1 A
- (B) 4 A, 2.5 A and 2 A
- (C) 4 A, 2.5 A and 1 A
- (D) 6 A, 4.5 A and 1.5 A

Correct Answer: None

Solution Analysis: At any junction, Kirchhoff's Current Law says that the total current entering the junction is equal to the total current leaving it. For the first junction, 5A enters. The currents leaving are 1A and I_1 . So, $5 = 1 + I_1 \Rightarrow I_1 = 5 - 1 = 4A$ For the second junction, I_1 enters. The currents leaving are 1.5A and I_2 . So, $I_1 = 1.5 + I_2$ Putting $I_1 = 4A$, $4 = 1.5 + I_2 \Rightarrow I_2 = 4 - 1.5 = 2.5A$ For the third junction, I_2 enters. The currents leaving are 1.5A and I_3 . So, $I_2 = 1.5 + I_3$ Putting $I_2 = 2.5A$, $2.5 = 1.5 + I_3 \Rightarrow I_3 = 2.5 - 1.5 = 1A$

Benchmark Problem 2: KCET 2026 (Q#1)

Given below are two statements: Statement-I: The resistivity of a conductor is independent of its temperature. Statement-II: The resistivity of a semiconductor decreases with increase in temperature. Select the correct option.

- (A) Both Statement I and Statement II are false
- (B) Both Statement I and Statement II are true
- (C) Statement I is true but Statement II is false
- (D) Statement I is false but Statement II is true

Correct Answer: None

Solution Analysis: Statement-I is false. For a conductor, especially a metal, resistivity depends on temperature. When temperature increases, the ions in the metal vibrate more strongly. This makes it harder for free electrons to move, so the resistivity increases. So, resistivity of a conductor is not independent of temperature. Statement-II is true. For a semiconductor, when temperature increases, more electrons get enough energy to move to the conduction band. This increases the number of charge carriers, so conductivity increases. Hence, resistivity decreases with increase in temperature. So the correct opti